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Pose-Invariant Facial Expression Recognition Based on 3D Face Reconstruction and Synthesis from a Single 2D Image

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Abstract— In this paper, a novel method is proposed for person-independent pose-invariant facial expression recognition based on 3D face reconstruction from only 2D frontal images in a training set. A 3D Facial Expression Generic Elastic Model (3D FE-GEM) is proposed to reconstruct an expression-invariant 3D model of each human face in the present database using only a single 2D frontal image with/without facial expressions. Then, for each 7-class of facial expressions in the database, a Feature Library Matrix (FLM) is created from yaw face poses by the rotating the 3D reconstructed models and extracting features in rotated face. Each FLM is subsequently rendered based on yaw angles of face poses. Before matching to the FLM, an initial estimate of yaw angles of face poses is obtained in the test face image using an automatic head pose estimation approach. Then, an array of the FLM is selected based on the estimated yaw angles for each class of facial expressions. Finally, the selected arrays from FLMs are compared with target image features by Support Vector Machine (SVM) classification. Favorable outcomes were acquired to handle pose in facial expression recognition on the available image based on the proposed method compared to several state-of-the-arts in pose-invariant facial expression recognition.

Keywords—Facial expression recognition; 3D shape recovery.

I. INTRODUCTION

Facial expression recognition is one of the most difficult and challenging tasks in computer vision because of unknown changes in the expression of human faces. Pose-invariant facial expression recognition from a 2D image in frontal pose is a more challenging problem than from facial expression recognition in frontal poses because less information for pose-invariant expression recognition is available, and also mainly due to a lack of suitable data. Most existing methods for facial expression recognition assume that the person in the images does not pose the head strongly and that the image shows a nearly frontal pose of the face. However, information in a single image is sometimes enough for pose-invariant expression recognition, and in many applications it is also useful to recognize a single pose image's facial expression.

Generally, facial expression recognition encompasses two main steps: feature extraction and classification [1]. The method used in feature extraction should give adequate features, otherwise it will be misclassified. The common facial

expression recognition approaches can be mostly categorized into four separate types: geometric-based, appearance-based, model-based and multi-view methods [1]. Geometric features contain information about the location and shape of facial features. Geometric features are more sensitive to noise and tracking errors than appearance features. An example of these methods is the graph-based methods [2-4], which utilize several facial components to make a representation of the face and process it.

The appearance-based approaches [5, 6] utilize image filters; either on the whole faces to construct global features, or several particular face-regions to construct local features, to extract the appearance changes in the face image. The performance of the appearance-based methods is excellent in a constrained environment but their performance degrades in environmental variation [7]. Appearance-based methods have been heavily employed in this domain with great success. Popular approaches are Gabor wavelets [8], Local Binary Patterns (LBP) operator [9, 10], Local Phase Quantization (LPQ) [11], and subspace learning methods, such as, Eigenfaces [12], Principal Component Analysis (PCA) [12], Linear Discriminant Analysis (LDA) [13], and etc.

As opposed to the appearance-based approaches, model-based methods describe the face by modeling its shape and its appearance. The downside of this approach is, that for the construction of the models, it is necessary to label the facial landmarks by hand. Therefore, having a big amount of training data implies a lot of manual labor. The examples of these methods are described in [14-16].

Multi-view expression recognition approaches extend frontal facial expression recognition methods in order to process expressive face images or video sequences at different pose angles of the face. In this context, Moore and Bowden [17] proposed an appearance based approach which is adopted by dividing images into sub-blocks coarsely aligned over the face. Then, feature vectors containing feature histograms were built from each sub-block by LBP. Finally, multiclass support vector machines were adopted to learn pose and pose dependent facial expression classifiers.

Recently, Heo and Savvides [18] proposed Generic Elastic Models (GEMs) as a novel impressive, fast and trustworthy 3D reconstruction technique from a single 2D image. In fact,

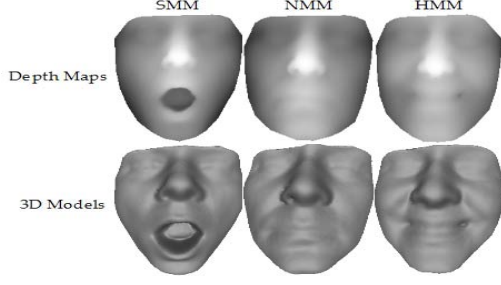


Fig. 1. Three generic models with depth maps, such as surprised mean model (SMM), neutral mean model (NMM) and happy mean model (HMM), are used for proposed 3D face reconstruction.

this method planned a 3D face model that could be efficiently produced by using generic depth models that could be elastically deformed to align with facial landmarks. Also, another extending of GEM methods that is expression-invariant can be found in [26-28].

In this paper, a new approach is proposed for person-independent pose-invariant facial expression recognition based on 3D reconstruction and synthesized the face. Accordingly, a 3D model is initially reconstructed from 2D frontal face images with facial expression. To reconstruct a 3D model from each human frontal face with facial expression, a Facial Expression Generic Elastic Model (FE-GEM) is proposed. The FE-GEM method for facial expression invariant 3D face reconstruction is the extension of GEM [18] to resolve the drawback of handling facial expression in 3D face reconstruction. Then, each 3D reconstructed face in the gallery is synthesized to all possible views and a Feature Library Matrix (FLM) is generated based on yaw angles of face pose for each 7-class. On the other hand, automatic head pose estimation by the eye-based normalization approach is used to extract the yaw angles of face poses. Therefore, for each 7-class, an array of FLM is selected based on yaw angles of face pose which are estimated from automatic head pose estimation. Finally, the facial expression recognition is performed by Support Vector Machine (SVM) [19] between the SLMs selected arrays and target images features.

This paper is organized as follows: Section II describes the 3D face modeling method from a single frontal face image. In section III, the generation manner of FLM and training method for each human face in the database are proposed for pose-invariant facial expression recognition. Experimental evaluations are given in section IV and conclusions are presented in section V.

II. CONSTRUCTING A 3D DATABASE

In order to construct a 3D database from an arbitrary frontal image with/without facial expression, a Facial Expression Generic Elastic Model (FE-GEM) [29-31], which is the extended GEM [18], was the method used due to its facial expression invariant 3D face reconstruction. In this section, initially, the FE-GEM method is proposed for facial expression invariant 3D face reconstruction from a single frontal face image. Then, constructing manner of 3D database from 2D images with facial expression is discussed.

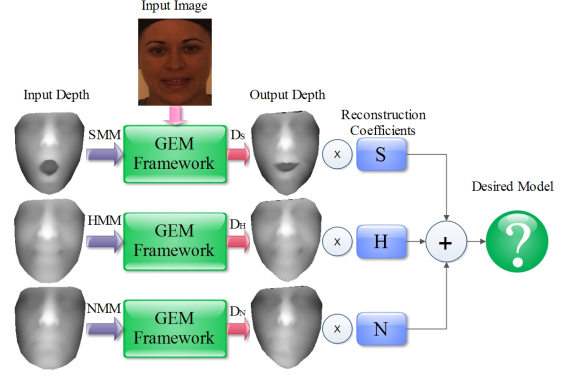


Fig. 2. Generation the desired model by mixture of facial expression models.

A. Facial expression generic elastic model

In this section, facial expression depth alterations in 3D human face models are examined and the method is presented for facial expression invariant 3D reconstruction from only frontal images. By investigating and analyzing the 3D human face with facial expression by Zhao et al. [20], the conclusion was obtained that maximum depth variation of neutral models was related to happy expression models and surprised expression models. Therefore, three models were employed to be utilized in the GEM framework. Fig. 1 shows three models including Surprised Mean Model (SMM), Neutral Mean Model (NMM) and Happy Mean Model (HMM), which are generated from the Bosphorus 3D database [21] by the proposed method for generating the generic model in [18].

By employing three generic models in the GEM framework and achieving three GEM models from input images, the desired model can be generated with a mixture of these three new models according to the similarity coefficients. An illustration of the problem is shown in Fig. 2. Therefore, the desired model is achieved as:

$$D = S \times D_s + N \times D_n + H \times D_h \quad (1)$$

where D is the desired depth model, D_s is surprised output depth model of GEM (Fig. 2), D_n is neutral output depth model of GEM, D_h is happy output depth model of GEM and S , N and H are surprised, neutral and happy coefficients, respectively.

By investigating the 2D face image and corresponding depth, reconstruction coefficients were obtained by computing distance ratio (DR) in 2D face input images as:

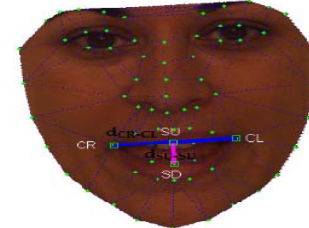


Fig. 3. 72 landmarking schemes used by CLM [22]. SU is stomion upper landmark, SD is stomion down landmark, CR is cheek right landmark, CL is cheek left landmark, d_{CR-CL} is distance between CR and CL, d_{SU-SD} is distance between SD and SU.



Fig. 4. Examples of 3D face reconstruction from a frontal image by FE-GEMs achieved from the images of famous people downloaded from the internet.

$$DR = \frac{d_{CR-CL}}{d_{SU-SD}} \quad (2)$$

where d_{CR-CL} is the distance between Cheek Right (CR) landmark and Cheek Left (CL) landmark, d_{SU-SD} is the distance between stomion upper landmark (SU) and stomion down landmark (SD), as shown in Fig. 3.

Similarity parameter (SP) is written as:

$$SP_i = DR_i - DR, i = S, N, H \quad (3)$$

where DR_i are the distance ratio in surprise mean model ($i=S$), neutral mean model ($i=N$) and happy mean model ($i=H$), and DR is the distance ratio in 2D face input image.

Thus, the reconstruction coefficient was obtained as:

$$S = \frac{SP_S}{SP_S + SP_N + SP_H} \quad (4)$$

$$N = \frac{SP_N}{SP_S + SP_N + SP_H} \quad (5)$$

$$H = \frac{SP_H}{SP_S + SP_N + SP_H} \quad (6)$$

where $S+N+H=1$, S , N , H can be positive or negative.

Examples of facial expression invariant 3D reconstruction of five images of famous people which were downloaded from the Internet using the FE-GEM method are also shown in Fig. 4. In the figure, the first column images contain image of famous people, the second column images contain the estimated 3D shapes and the third to final column images contain 3D texture models in different new poses from -75 degrees up to +75 degrees. These 3D models were generated from their corresponding input images in the first column.

B. 3D face database without limiting facial expression

In order to construct a 3D database from an arbitrary frontal image with/without facial expression, the FE-GEM method

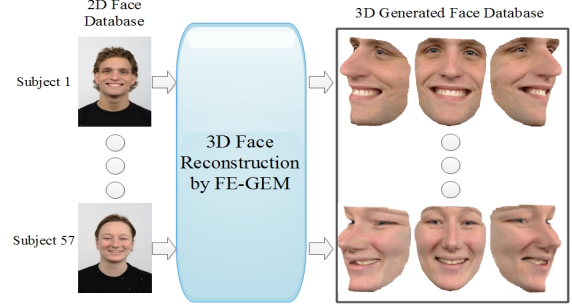


Fig. 5. Visual illustration of constructing the 3D face database from frontal face images.

was used. To construct the 3D database for each facial expression, one 2D frontal image was registered for each facial expression per person. Therefore, 2D images for constructing the 3D database included one frontal image with any facial expression. In this work, 57 people from the Radboud database [23] were used in which all images are frontal face images with seven facial expressions.

Therefore, 57 frontal face images were registered for each facial expression to construct a 3D database from these images. Visual illustration of the constructing a 3D face database from frontal face images from a database was registered, as shown in Fig. 5. Based on each 2D frontal face image with/without facial expression which was registered from a 2D database, 3D faces were reconstructed and a 3D face database was made.

III. POSE-INVARIANT FACIAL EXPRESSION RECOGNITION

In this section, the feature extraction method is proposed from a 3D database by Local Binary Pattern [9, 10] (LBP) for pose-invariant facial expression recognition. Then, the method for pose-invariant facial expression recognition is represented.

A. Feature Extraction by LBP

Visual illustration of the proposed method for extracting the feature by LBP [9, 10] for pose-invariant facial expression recognition is shown in Fig. 6. Based on the proposed method, the process can be summarized as follows:

1) Input: 7 sets of the 3D faces database based on 7-classes of traditional facial expression. These seven classes include: neutral, sad, angry, fearful, disgusted, happy and surprised.

2) For each 7-class, 3D faces were synthesized based on face pose and possible views in all poses of the face were extracted with steps of 5 ($s=5$) degrees on either side at yaw direction in the pose matrix $P_{i,Y}$. In fact, arrays of each $P_{i,Y}$ matrix for each 7-class were arranged based on the yaw angle of face pose from the corresponding class, where i indicates the number of class ($i=1$ neutral, $i=2$ sad, $i=3$ angry, $i=4$ fearful, $i=5$ disgusted, $i=6$ happy and $i=7$ surprised) and Y is the number of array in the yaw dimension of $P_{i,Y}$ matrix.

3) The feature vectors were extracted based on Fig. 7 (example of feature extraction for each image is demonstrated in Fig. 7) from each array of pose matrix $P_{i,Y}$ and by the LBP method. Then, for each specific pose yaw, an array of Feature Library Matrix (FLM) $A_{i,Y}$ was created which was a feature

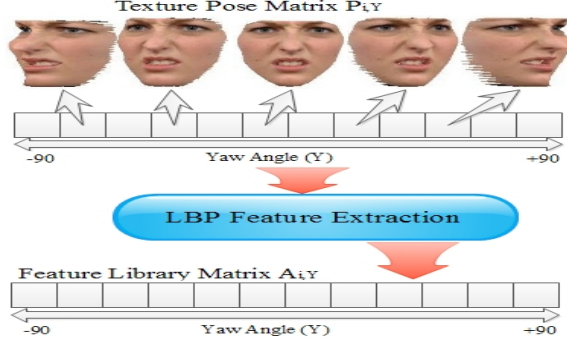


Fig. 6. Visual illustration of creating FLM from the present 3D database. In this figure, an example of creating the $A_{6,Y}$ for happy expressions are given.

set of created views by synthesizing 3D reconstructed models. However, for each 7-class, a FLM $A_{i,Y}$ was created based on the yaw angle of face pose.

5) Output: FLMs $A_{i,Y} = [A_{1,Y}, A_{2,Y}, A_{3,Y}, A_{4,Y}, A_{5,Y}, A_{6,Y}, A_{7,Y}]$ where i is the number of classes in the database.

To extract a high dimensional feature, a local histogram of LBP for the face was applied according to Fig. 7. Then, a low dimensional feature was generated by Principal Component Analysis (PCA) from a high dimensional feature.

B. Facial Expression Recognition System

The visual illustration of the person-independent pose-invariant facial expression recognition system proposed in this paper is shown in Fig. 8. The proposed system operated in two offline and online stages. In the offline stage, a single frontal face image of each person with seven facial expressions was registered from an available 2D database. Then, for each registered image, 3D face reconstruction was performed by the FE-GEM to produce a 3D face database based on 7-classes of facial expression. Afterwards, feature extraction was performed and FLM was created for each facial expressions.

In the online stage, pre-processing and face detection were performed for each test image by CLM landmark localization [22]. Then, yaw angles of the face were estimated by the eye-based normalization head yaw estimation method. Afterwards, for each 7-class, an array ($A_{i,Y=yaw}$) of FLM was selected based on yaw estimated angles. Finally, facial expression recognition was performed by SVM [19] (linear) between the target image features and selected FLMs $A_{i,Y=yaw} = [A_{1,Y=yaw}, A_{2,Y=yaw}, A_{3,Y=yaw}, A_{4,Y=yaw}, A_{5,Y=yaw}, A_{6,Y=yaw}, A_{7,Y=yaw}]$.

IV. EXPERIMENTS

In this section, person-independent facial expression recognition was performed based on the proposed method in the previous section. For this experiment, a subset of the Radboud database with seven different facial expressions at five different poses (-90, -45, 0, +45 and +90 degrees) were used for testing images. The Radboud faces database [23] included 20 Caucasian male adults, 19 Caucasian female adults, 18 Moroccan male adults, 6 Caucasian female children and 4 Caucasian male children and face images of each person

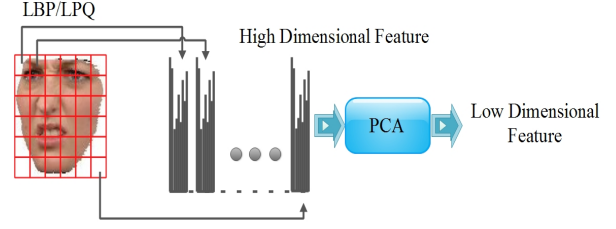


Fig. 7. Visual illustration of feature extraction by LBP/LPQ from face images using Principal Component Analysis (PCA).

were captured with three different gaze directions. The seven expressions included neutral, sad, angry, fearful, disgusted, happy and surprised. In this part of the work, a subset of 57 subjects were utilized which included 20 Caucasian males, 19 Caucasian females and 18 Moroccan males in seven facial expressions and three different gaze directions. Examples of facial expressions and different gaze directions in the Radboud faces database at different face poses are shown in Fig. 9. Therefore, there are 171 (57×3 , three gaze different direction) images from 57 people for each facial expression in each face pose. Hence, the total images utilized in this experiment for seven facial expressions are 1197 (171×7) for each face pose.

To evaluate the generalization performance of novel subjects, a 10-fold cross-validation testing scheme was adopted in the present experiments for person-independent facial expression recognition. The confusion matrices of 6-class and 7-class recognition at ± 45 degrees of face pose by the proposed method are shown in Tables I and II, respectively. Also, the confusion matrices of 6-class and 7-class recognition at ± 90 degrees of face pose by the proposed method are shown in Tables III and IV, respectively.

Moreover, to evaluate further accuracy of the present method for pose-invariant facial expression recognition from static images in the gallery, the obtained results were compared with seven feature extraction methods for tests based on using the SVM (linear) classifier to classify the features:

- 1- Local phase quantization (LPQ) [11] + proposed method. In this method, instead of the LBP operator the LPQ operator was applied to the image for feature extraction based on Fig. 7.
- 2- Local Gabor Binary Patterns (LGBP) [17] + proposed method. In this method, instead of the LBP operator the LGBP operator was applied to the image for feature

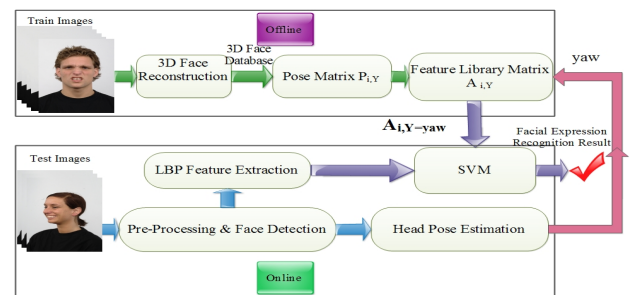


Fig. 8. Visual illustration of the facial expression recognition from a static face image and 3D face synthesis.

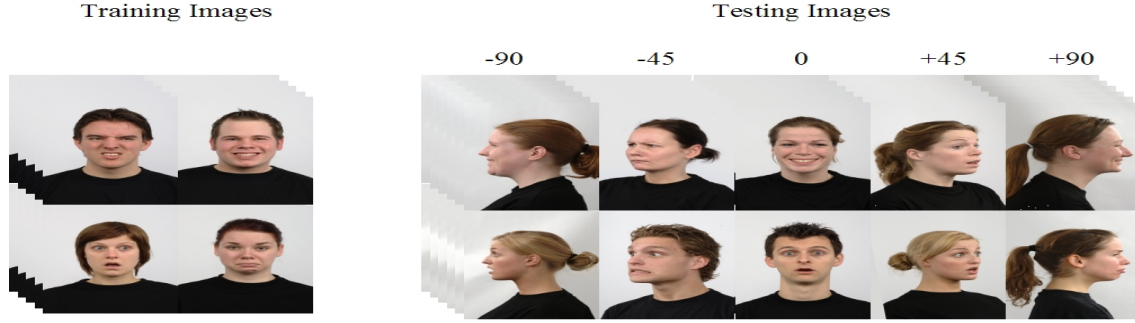


Fig. 9. Sample of the face images utilized for the present experiments with seven facial expressions at five face poses.

extraction based on Fig. 7.

- 3- Gabor filters bank [25] + proposed method. In this method, instead of the LBP operator feature vectors were created by applying the Gabor filter bank to the image. To extract the Gabor feature, a Gabor filter bank with a size of 40 (5*8 with 8 directions and 5 magnitudes) was applied to each image of the pose matrix. Then, a feature vector was created from the entire 40 magnitude of 40 Gabor filter bank which was low dimensional by the downsampling method.
- 4- Downsampling [25] + proposed method of images with size 12×10. In this method, instead of the LBP operator the feature vectors were created by the downsampling process on only texture images of which the length of feature vectors are 120.
- 5- LGBP+ sub-blocks method [17]. This method was implemented based on as presented in [17]
- 6- Multi-Scale Local Binary Patterns (LBP^{ms}) + sub-

blocks method. This method was implement based on as presented in [17]

The proposed facial expression recognition method seems to outperform the other implemented approaches. These measurement results are shown in Table V. Table V shows the best recognition rates of the proposed methods, in comparison with other methods. As is obvious from the results, the performance of the present method for person-independent pose-invariant facial expression recognition was better than the state-of-the-art methods.

V. CONCLUSION

In this paper, a new combined approach for person-independent pose-invariant facial expression recognition from only 2D frontal images in the gallery was presented. Accordingly, a 3D model was initially reconstructed from training frontal face images with/without facial expression by the FE-GEM method. Then, different views were created by

TABLE I. CONFUSION MATRIX OF 6-CLASS FACIAL EXPRESSION RECOGNITION USING PROPOSED METHOD AT ± 45 DEGREES OF FACE POSE

%	happy	disgusted	fearful	surprised	angry	sad
happy	88.7	1.1	2.6	0	4.2	3.4
disgusted	1.3	86.1	7.9	2.5	1.4	0.8
fearful	2.1	7.3	87.6	3.0	0	0
surprised	0	4.4	4.1	91.5	0	0
angry	1.9	0.5	0	0	84.8	12.8
sad	2.2	1.2	0	0	10.9	85.7

TABLE II. CONFUSION MATRIX OF 7-CLASS FACIAL EXPRESSION RECOGNITION USING PROPOSED METHOD AT ± 45 DEGREES OF FACE POSE

%	neutral	happy	disgusted	fearful	surprised	angry	sad
neutral	81.3	2.4	1.1	0.5	0	8.3	6.4
happy	1.7	87	1.1	2.6	0	4.2	3.4
disgusted	1.4	1.3	85.1	7.9	2.5	1.1	0.7
fearful	0.7	1.9	6.8	87.6	3.0	0	0
surprised	0	0	4.4	4.1	91.5	0	0
angry	8.3	1.6	0.5	0	0	78.0	11.6
sad	9.8	0.8	0	0	0	10.0	79.4

TABLE III. CONFUSION MATRIX OF 6-CLASS FACIAL EXPRESSION RECOGNITION USING PROPOSED METHOD AT ± 90 DEGREES OF FACE POSE

%	happy	disgusted	fearful	surprised	angry	sad
happy	89.4	2.9	3.1	0	2.5	2.1
disgusted	0.3	87	9.4	2.9	0.3	0.4
fearful	0.9	5.1	91.2	2.8	0	0
surprised	0	1.5	2.8	95.7	0	0
angry	0	0	0	0	77.9	23.1
sad	0	0	0	0	20.8	79.2

TABLE IV. CONFUSION MATRIX OF 7-CLASS FACIAL EXPRESSION RECOGNITION USING PROPOSED METHOD AT ± 90 DEGREES OF FACE POSE

%	neutral	happy	disgusted	fearful	surprised	angry	sad
neutral	76.8	4.7	1.3	0	0	9.8	7.4
happy	2.8	87.2	2.8	3.1	0	2.2	1.9
disgusted	0.4	0.3	86.9	9.1	2.9	0.3	0.4
fearful	0	0.9	5.1	91.2	2.8	0	0
surprised	0	0	1.5	2.8	95.7	0	0
angry	12.5	0	0	0	0	73.2	14.3
sad	13.7	0	0	0	0	12.4	73.9

synthesizing the 3D reconstructed faces and these created views of faces were used in the pose-invariant facial expression recognition system based on the proposed method by making a feature library matrix from each 7-class.

The proposed method was tested on the available image in order to perform person-independent pose-invariant facial expression recognition. Also, the obtained results showed which one could handle the pose of face based on the proposed method to perform pose-invariant facial expression recognition. It was demonstrated that performance of the proposed method for pose-invariant facial expression recognition was better than similar approaches and experimental results of the proposed method were compared with classical, as well as state-of-the-art methods.

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TABLE V. RECOGNITION RATE ON THE RADBOUD FACE DATABASE FOR SEVERAL METHODS.

Method	6-class at ± 45	7-class at ± 45	6-class at ± 90	7-class at ± 90
Proposed method	87.4	84.2	86.7	83.5
LPQ + proposed method	85.8	83.0	85.5	82.8
LGBP + proposed method	85.7	84.2	84.8	83.6
Gabor + proposed method	83.1	81.4	82.7	79.4
Downsampling + proposed method	80.6	78.0	80.1	77.2
LGBP + sub-block	68.3	61.7	57.8	53.1
LBP^{ms} + sub-block	63.7	60.3	55.9	52.4

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